

Estimates of genetic parameters and breeding values from western larch open-pollinated families using marker-based relationship

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Received: 23 May 2013 / Revised: 16 September 2013 / Accepted: 8 October 2013 / Published online: 8 November 2013
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Abstract The availability and affordability of genetic markers made it possible to estimate quantitative genetic parameters without mating designs' structured pedigree. Here, we compared 4-year height's heritability and individuals' breeding values for a western larch common-garden population of 1,418 offspring representing 15 open-pollinated families from a 41-clone seed orchard using (a) classical pedigree models such as half- and full-sib families and (b) a molecular marker-based pedigree-free model using four pair-wise relationship estimation methods using eight informative SSR markers. The results highlighted the commonly observed inflated estimates of genetic parameters often obtained from half-sib analyses, as well as demonstrating some of the full-sib analyses' caveats. The pedigree reconstruction permitted the identification of selfed individuals, thus allowing evaluating the impact of selfing on marker-based genetic parameter estimation. The results demonstrated the utility of marker-based methods as an alternative to the classical pedigree-based approaches. Unlike the pedigree-based methods, the marker-based approach allowed better partitioning the variance components as well as separating the non-additive and additive genetic variance. The theoretical underpinning of the marker-based approach was discussed.

Keywords Pedigree-free model · Pedigree reconstruction · Marker-based pair-wise relationship · Genetic parameter estimates · Open-pollinated families · SSR · *Larix occidentalis*

Introduction

In traditional tree improvement programs, the selection of elite genotypes is often concluded after breeding, testing, and the ranking of parents and their offspring's breeding values (Namkoong et al. 1988). This process is accomplished using the intra-class correlation approach (Falconer and Mackay 1996), and thus, the generation of structured pedigree such as those produced from mating designs is an essential prerequisite (White 1987). Structured pedigree is not only essential for quantitative genetic parameter estimation but also important for the management of co-ancestry, whereby inbreeding can be strategically managed (Lindgren and Mullin 1997; Hallander and Waldmann 2009). As breeding programs advance, the buildup of co-ancestry is unavoidable, unless gain is intentionally sacrificed, a decision that compromises the improvement enterprise's main goal. Quantitative genetic data analyses such as the best linear unbiased predictor (BLUP) account for inbreeding buildup as they consider the known genealogies among parents and their offspring (Xu 2003). However, analyses focusing on the base population often overlook inbreeding and result in biased estimates of genetic parameters because any historical inbreeding is often unknown. A pedigree-based relationship (*A*-matrix; average numerator relationship matrix) is often used in these analyses (Henderson 1976); however, while useful, it has its limitations specifically in complicated multi-generational pedigree. Even for simple pedigree such as half-sib (HS) and full-sib (FS) families, pedigree-based relationship methods do not differentiate among members of

Communicated by D. Neale

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a HS or FS family; moreover, in each case, the offspring are given a “global” relationship estimate without adjustment for the Mendelian sampling term (i.e., variation about the average relatedness among members of a full-sib or a half-sib family that cannot be determined by traditional pedigree methods or records).

Marker-based pedigree reconstruction of offspring produced through natural matings has proven to be helpful in avoiding expensive controlled mating, simplifying testing, and providing better genetic diversity control at the selection phase (El-Kassaby and Lstibůrek 2009; El-Kassaby et al. 2011). Similar to mating designs, however, the pedigree generated after reconstruction also relies on the expected coefficients of genetic covariances that are not adjusted for the Mendelian sampling term (Avendaño et al. 2005). The availability and affordability of DNA genotyping and/or next-generation sequencing data provided a powerful tool allowing greater relationship differentiation even among related individuals including members within each class of relatedness (e.g., full- or half-sib) as the relationship can be traced back to multiple generations (Powell et al. 2010). Comparisons between marker-based and traditional pedigree relationship estimates indicated that the former provide a continuum of relationships, as opposed to the discrete classes of the latter, and thus, it is expected to reflect the real contemporary and historic genealogy among the tested individuals (Fig. 1).

The use of marker-based relationship estimates frees breeders from the cumbersome traditional process of tracking pedigree that in this case could be considered as a limitation rather than an advantage. Traditional breeding can be substantially simplified

through the use of marker-based relationship estimates, thus offering independence from structured pedigree without sacrificing the accuracy or precision of the estimated genetic parameters. The use of marker-based relationship estimates (i.e., pedigree-free approach) is expected not only to simplify tree breeding but also to create new opportunities that could not be contemplated through traditional tree improvement approaches, specifically in natural populations and undomesticated species. A limitation of the marker-based relationship estimators is the lack of prior knowledge about the base population's (parental generation) allelic frequencies, for which a proxy is given by the allelic frequencies of the population under study (Oliehoek et al. 2006) or of a selected subset of unrelated individuals from the analyzed population (Nomura 2008). It should be pointed out that in traditional tree breeding, prior knowledge about relatedness in the base population is also lacking, and individuals were often assumed to be unrelated and non-inbred (Namkoong et al. 1988; White 1987). Moreover, the testing population's estimated genetic parameters are often biased upwardly due to cryptic relatedness in the population under study (Askew and El-Kassaby 1994) which can be effectively revealed through pedigree reconstruction (El-Kassaby et al. 2011). However, pedigree reconstruction is also subject to pedigree-assignment errors, and these can only be overcome by the reliance on marker-based relationship estimates.

Here we used a western larch (*Larix occidentalis*) common-garden experiment, originally established with replicated open-pollinated families, and full-sib (pair-cross)

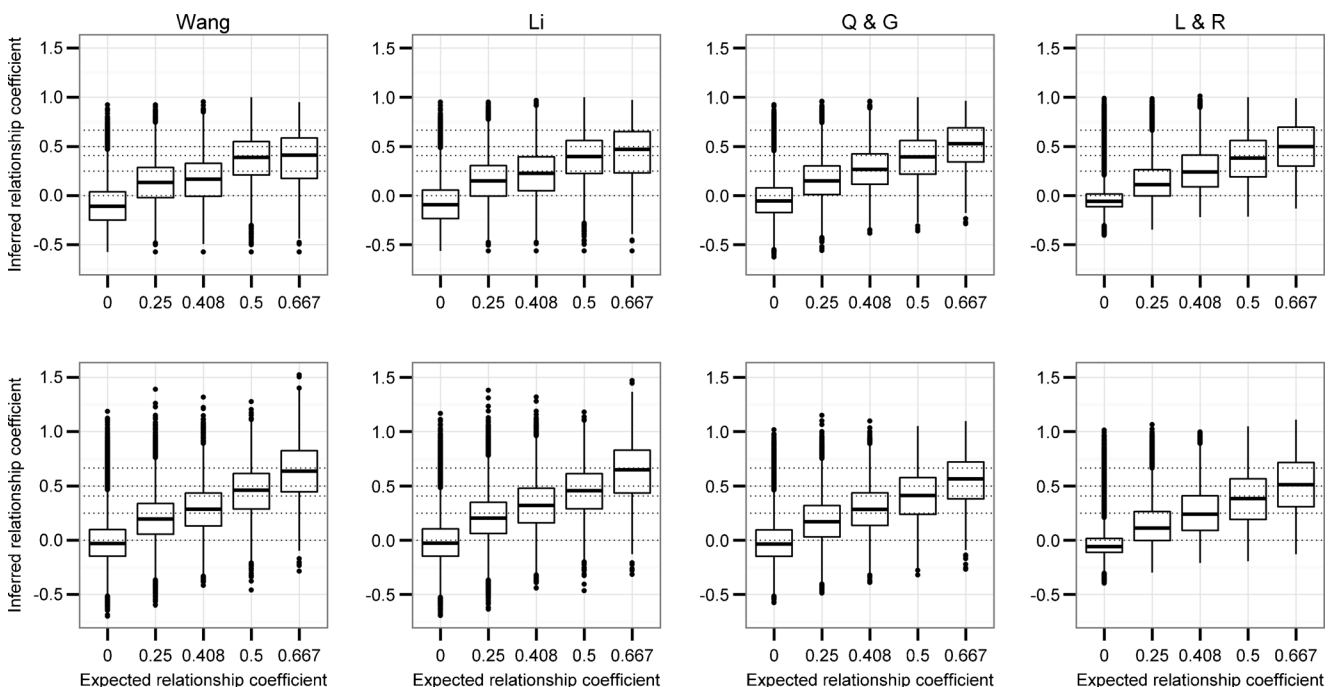


Fig. 1 Box plots portraying the distribution of marker-based relationship coefficient estimates (upper row) and those after nearPD function application (lower row) within each class of expected relatedness

(unrelated (0.0), half- (0.25) and full-sib (0.5), self half-sib (0.408), and self full-sib (0.667)). L & R Lynch and Ritland (1999), Q & G Queller and Goodnight (1989), Wang (2002), Li et al. (1993)

families were later assembled using simple sequence repeat (SSR) markers for pedigree reconstruction (El-Kassaby et al. 2011) to estimate breeding values and heritability using marker-based relationship estimates (i.e., pedigree free). Moreover, the pedigree reconstruction has identified the presence of 113 selfed individuals, thus providing the opportunity to investigate the effect of selfing on the marker-based estimates. The study's main focus is to evaluate the marker-based relationship in estimating the quantitative genetic parameters of this population and compare them to those produced using traditional pedigree categories.

Material and methods

Study population and pedigree reconstruction

Phenotypic (age 4-year height, average=74.6 cm) and genotypic (10 SSRs) data for 1,418 randomly selected western larch individuals growing in a common-garden trial (El-Kassaby et al. 2011) were used in this study. Briefly, the trial was established as replicated open-pollinated (OP) families using seeds from 15 of a 41-parent western larch seed orchard. The randomly selected individuals (1,418 out of 6,000) were fingerprinted, and following partial pedigree reconstruction (using the CERVUS software (Kalinowski et al. 2007)), their paternal parents were determined and 349 FS families were assembled within the studied 15 OP families (El-Kassaby et al. 2011). The pedigree reconstruction identified 113 individuals as selfs, thus providing an opportunity to assess the effect of selfing on the performance of marker-based quantitative genetic parameter estimates (see below). The identity of each individual was retained as either a member of the original 15 putative HS families (range 37–148 individuals) or the newly assigned FS families (range 1–34), thus permitting quantitative genetic data analyses using traditional pedigree-based HS and FS models. The paternity analysis utilized 10 informative SSR loci (number of alleles/locus average=8.5 (range 4–14), average observed heterozygosity=0.679 (range 0.488–0.902), average expected heterozygosity=0.716 (range 0.528–0.877), average polymorphic information content (PIC, the relative measure of a marker's informativeness that is dependent on its number of alleles and their relative frequency in the population)=0.646 (range 0.420–0.842)) and resulted in 98 % of the genotyped offspring assigned to within-orchard pollen parents. A genotyping error rate of 0.01 across the 10 loci was assumed, and paternity assignment was based on 10,000 simulations, with the 41 parents as candidate fathers. The log-likelihood (LOD) score, the likelihood that the candidate parent is the true parent divided by the likelihood that the candidate parent is not the true parent, was calculated for each putative parent. The delta score, the difference in LOD scores

of the two most likely candidate parents, was used as a criterion for assignment of parentage at the 95 % level of confidence in our analysis (El-Kassaby et al. 2011).

Marker-based relationship estimation methods

Marker data were used to estimate pair-wise relationship coefficients among the fingerprinted individuals and the generated matrix (*G*-matrix) were included in the quantitative genetic analyses in place of the pedigree-based matrix needed for restricted maximum likelihood (REML)–BLUP (Mrode 1996). Four marker-based pair-wise relationship estimators were used (Queller and Goodnight 1989; Li et al. 1993; Lynch and Ritland 1999; Wang 2002), implemented in SPaGeDi v. 1.3 software (Hardy and Vekemans 2002).

The quality of molecular marker-based relationship matrices depends on both the number and informativeness of the markers used (Rosenberg et al. 2003). To prevent bias of relationship coefficient estimates, only 8 out of the 10 markers were used in estimating pair-wise relationship coefficients (only loci with PIC values >0.5 were retained (Botstein et al. 1980)). While the methods used do not estimate inbreeding coefficients, diagonal elements were set as 1 (Yu et al. 2006; Li et al. 2011). Throughout the study, two scenarios were compared: (1) total population ($N=1,418$) and (2) total population after the removal of selfs ($N=1,305$). In this context, the studied population is divided to either unrelated (UN), HS and FS, self half-sib (self HS), and self full-sib (self FS) (see Table 1).

Often, molecular marker-based pair-wise relationship matrices are not positive definite due to internal inconsistency resulting from the lack of adequate markers, genotyping errors, or missing values; thus, we used the “nearPD” function in R package “Matrix” to compute the nearest positive definite matrix to an approximate one (Knol and Tenberge 1989; Cheng and Higham 1998; Higham 2002). The algorithm iteratively searches for a positive definite solution for the arbitrary symmetric matrix under study $A \in \mathbb{R}^{n \times n}$ by minimizing the Frobenius distance:

$$\gamma(A) = \min\{\|A - X\|_F : X = X^T, X \geq 0, \text{Diag}(X) = e\} \quad (1)$$

where for symmetric matrices X and Y , $X \geq Y$ denotes that $X - Y$ is positive semidefinite, $\text{Diag}(X)$ is the vector of diagonal elements of X , e is the vector of ones, and the Frobenius norm $\|X\|_F = \text{trace}(X^T X)^{1/2}$. As $\mathbb{R}^{n \times n}$ is a Hilbert space with inner product $\langle X, Y \rangle = \text{trace}(X^T Y)$ and constraints in Eq. 6 are closed convex sets, Eq. 6 has a unique solution. The algorithm wrapped in the nearPD function provides several favorable features: simple implementation requiring only matrix additions and computing of eigendecomposition; can take advantage of the property that if $a_{ii} \geq 1$ for all i and A has many negative eigenvalues, then the solution has at least

Table 1 Mean and variances (in brackets) of marker-based relationship coefficients within each class of relatedness (original relationship coefficients and after nearPD function application) with their expected coefficients and % representation

	Relatedness class	Selfing excluded			Selfing included				
		UN	HS	FS	UN	HS	Self HS	FS	Self FS
	Expected coefficients	0.0	0.25	0.5	0.0	0.25	0.408	0.5	0.667
	Class representation (%)	(81.8)	(17.6)	(0.6)	(82.8)	(14.9)	(1.7)	(0.5)	(0.06)
Original	L & R	−0.035 (0.014)	0.145 (0.037)	0.380 (0.062)	−0.035 (0.015)	0.145 (0.037)	0.260 (0.048)	0.381 (0.062)	0.488 (0.071)
	Q & G	−0.038 (0.033)	0.159 (0.043)	0.385 (0.057)	−0.038 (0.033)	0.161 (0.042)	0.270 (0.048)	0.386 (0.057)	0.510 (0.058)
	W	−0.085 (0.040)	0.128 (0.047)	0.370 (0.060)	−0.100 (0.042)	0.131 (0.048)	0.162 (0.057)	0.372 (0.061)	0.384 (0.081)
	L	−0.073 (0.041)	0.146 (0.049)	0.385 (0.061)	−0.084 (0.043)	0.148 (0.049)	0.219 (0.061)	0.386 (0.061)	0.435 (0.084)
nearPD	L & R	−0.033 (0.014)	0.146 (0.037)	0.384 (0.064)	−0.034 (0.015)	0.146 (0.037)	0.261 (0.047)	0.386 (0.064)	0.505 (0.075)
	Q & G	−0.017 (0.032)	0.178 (0.041)	0.405 (0.055)	−0.018 (0.032)	0.179 (0.042)	0.288 (0.046)	0.406 (0.056)	0.546 (0.061)
	W	−0.014 (0.031)	0.193 (0.041)	0.448 (0.059)	−0.018 (0.032)	0.199 (0.041)	0.286 (0.050)	0.449 (0.059)	0.635 (0.085)
	L	−0.011 (0.034)	0.206 (0.043)	0.446 (0.056)	−0.014 (0.034)	0.207 (0.043)	0.319 (0.053)	0.447 (0.056)	0.637 (0.087)

L & R Lynch and Ritland (1999), *Q & G* Queller and Goodnight (1989), *W* Wang (2002), *L* Li et al. (1993), *UN* unrelated, *HS* half-sib, *FS* full-sib, *Self HS* self half-sib, *Self FS* self full-sib

many zero eigenvalues (so is of low rank); it is readily adapted to solve the problem with additional constraints that require X to belong to a convex set, such as a constraint that holds any set of elements of X fixed.

Estimation of genetic parameters and breeding values

The variance components, heritability, and breeding values were estimated with the REML animal model (Kruuk 2004) in ASReml v. 3.0 (Gilmour et al. 2002) as follows:

$$y = X\beta + Zp + Zf + Zu + e \quad (2)$$

where X and Z are incidence matrices relating fixed effects (intercept) in vector β and random effects in vector p (plot effects), vector f (full-sib family (specific combining ability) effect assumed=1/4 dominance variance) was replaced by vector d (dominance effect) obtained after the utilization of the dominance relationship matrix estimated in R package “nadiv” (Wolak 2012) ignoring the inbreeding (de Boer and Hoeschele 1993), and vector u (individual breeding values) to measurements in vector y , and e is the vector of residuals following $E \sim N(0, \sigma_e^2)$ (note: this X matrix is different from that mentioned in the nearPD function stated above). The vector of breeding values follows the relationship $\text{Var}(u) = A \times \sigma_a^2$ where σ_a^2 is the additive genetic variance and A is the average numerator relationship matrix in the FS model. Similarly, half-sib analysis was performed using original pedigree, and this would eliminate the full-sib term (Zf) from Eq. 7.

In the pedigree-free model, the A matrix is substituted by the marker-based pair-wise relationship matrix. The absence of pedigree does not allow the creation of a full-sib family structure, and thus, the model is simplified as follows:

$$y = X\beta + Zp + Zu + e \quad (3)$$

where description of all terms in the model is the same as above with the exception of the A matrix which in this case represents the marker-based relationship matrix (G -matrix)

The Lynch and Ritland (1999) and Wang (2002) methods allow estimating the fraternity coefficients which can be used to estimate the dominance variance component when the FS family structure is absent (Ovaskainen et al. 2008). We modified the fraternity matrices based on molecular marker information in the same way as the marker-based relationship matrix to reach a positive definite solution. The variance components were estimated in the linear mixed model as follows:

$$y = X\beta + Zp + Zd + Zu + e \quad (4)$$

where X and Z are incidence matrices relating fixed effects in vector β and random effects in vector p (plot effects), vector d (dominance effect), and vector u (individual breeding values) to measurements in vector y , and e is the vector of residuals following $E \sim N(0, \sigma_e^2)$. The vector of breeding values follows $\text{Var}(u) = G \times \sigma_a^2$ where G is the marker-based relationship matrix. Similarly, the vector of dominance effects is $\text{Var}(d) = D \times \sigma_d^2$ where σ_d^2 is the dominance variance and D is the matrix of fraternity coefficients.

Narrow-sense heritability estimates reflect the linear mixed model structures described above and are obtained as follows:

$$\hat{h}^2 = \frac{\hat{\sigma}_a^2}{\hat{\sigma}_a^2 + \hat{\sigma}_p^2 + \hat{\sigma}_e^2} \quad (5)$$

for model 3 and

$$\hat{h}^2 = \frac{\hat{\sigma}_a^2}{\hat{\sigma}_a^2 + \hat{\sigma}_d^2 + \hat{\sigma}_p^2 + \hat{\sigma}_e^2} \quad (6)$$

for models 2 and 4, where σ_a^2 , σ_d^2 , σ_p^2 , and σ_e^2 are the additive genetic variance, dominance variance, plot variance, and residual variance, respectively. The predicted breeding values' accuracies were estimated as follows:

$$r = \sqrt{1 - \frac{\text{PEV}}{(1 + F_i)\hat{\sigma}_a^2}} \quad (7)$$

where PEV is the prediction error variance and F_i is the inbreeding coefficient of the i th individual (Gilmour et al. 1995). To obtain comparable accuracy estimates across models, the variance components were set to those estimated in the equivalent FS model.

Results and discussion

Assuming no pedigree reconstruction errors, when the original data were used (i.e., before the implementation of the nearPD function), the results from all the four marker-based pair-wise relationship methods, including and excluding selfs, showed that the average pair-wise relationship coefficients for the putatively unrelated individuals were consistently close to 0 indicating the lack of relatedness among parents in the studied population (Fig. 1 and Table 1). For the HS and FS classes, including or excluding selfing, all pair-wise relationship estimates were below the expected value of 0.25 and 0.50, respectively, with slight variation among the four marker-based methods (Fig. 1, upper row and Table 1). The availability of the pedigree in the studied population permitted parsing the selfs from both the HS and FS families and allowed the creation of two additional relationship classes, namely half- and full-sib selfing with their expected relationship values of 0.408 and 0.667, respectively (Askew and El-Kassaby 1994; Fig. 1 and Table 1). Similar to the HS and FS classes and across the four relationship estimation methods, the two newly created classes produced pair-wise relationship estimates below their expected value (Fig. 1, upper row and Table 1). It is noteworthy that the consideration of the half- and full-sib selfs in the original analyses did not affect the resulting relationship estimates, thus producing similar results (Fig. 1 and Table 1).

The marker-based estimators usually produced non-positive definite relationship matrices due to internal inconsistencies which in turn makes it difficult to implement in linear mixed models to infer genetic parameters (i.e., additive genetic variance, heritability) (Blonk et al. 2010). Such matrices have to be modified by the bending method to meet mixed model assumptions (Blonk et al. 2010); thus, we implemented the nearPD function to approximate the positive definite matrix required. The four relationship estimation methods produced slightly variable results, with Lynch and Ritland (1999) as a notable exception (Fig. 1, lower

row and Table 1). This is expected as Lynch and Ritland's method works best in unrelated populations (van de Castelee et al. 2001). On average, the implementation of the nearPD function produced pair-wise relationship estimates close to expected values (note Lynch and Ritland's as exception) (Fig. 1, lower row and Table 1). Generally, the unrelated, HS, self half-sib, FS, and self full-sib produced mean relationship values close to 0.000, 0.250, 0.408, 0.500, and 0.667, respectively, supporting the use of marker-based estimates as a reliable substitute to known full pedigree. This indicates that the observed pair-wise relationships are a reflection of the most recent (reflecting the known pedigree) rather than ancient relatedness, as supported by the observed lack of relatedness among the seed orchard parents (i.e., lack of common ancestors) (cf. Powell et al. 2010).

It should be stated that the marker-based estimators do not produce inbreeding coefficient estimates, and it is recommended that the diagonal elements of the relationship matrix be set to 1 (Yu et al. 2006). Such a modification, however, assumes no inbreeding in the population, a situation that is not met in the present study. The implementation of the nearPD function allows inferring the diagonal elements according to the structure of the off-diagonal elements (Knol and Tenberge 1989; Cheng and Higham 1998; Higham 2002). When the expected inbreeding coefficients between inbred (selfs) and non-inbred (all other individuals) are compared, the marker-based methods differed in their ability to distinguish between these two classes with Wang (2002) (0.2695 vs. 0.1699) and Li et al. (1993) (0.2290 vs. 0.1123) showing a clear separation, while Queller and Goodnight (1989) (0.0576 vs. 0.0397) and Lynch and Ritland (1999) (0.0497 vs. 0.0342) did not (Fig. 2).

Comparing components of variance across models (HS, FS, and pedigree-free marker-based relationship) after including and excluding selfs showed that (1) the HS model produced unrealistically high additive genetic variance estimates resulting in inflated narrow-sense heritability estimates due to the presence of cryptic relatedness (presence of full-sib families and selfs; Namkoong (1966); Squillace (1974); Askew and El-Kassaby (1994)); thus, this model will not be discussed further; (2) the FS model was the only pedigree-based model permitting estimating the non-additive variance component (dominance); (3) the inclusion of selfs in the FS model as well as in the marker-based model permitted accommodating the matrix of fraternity coefficients (Ovaskainen et al. 2008) and resulted in the production of substantially inflated estimates of dominance genetic variance (this could be the results of substantial unbalance in the FS family structure, the presence of different selfing proportions within the original HS families, the assumption of no inbreeding in the pedigree-based dominance matrix, or the presence of different levels of inbreeding depression); and (4) the pedigree-free model, while it more specifically estimates

the true relationships among individuals, does not allow the formation of “full-sib like” families required for estimating non-additive genetic variance unless the matrix of fraternity coefficients is estimated (Table 2). The ability to separate the non-additive (assumed dominance) from additive genetic effects is largely dependent on the amount of variation in fraternity coefficients not explained by variation in relationship coefficients (Ovaskainen et al. 2008). Our results demonstrated the ability of the fraternity matrix to fit part of the variance component only when selfing occurred, as it caused the introduction of additional variance that could not be captured using relationship coefficients. The essential part of the captured dominance variance was extracted from the residual variance, while both the additive genetic and plot variances contributed negligibly. It should be clearly stated that we only managed to parse out the non-additive component in cases where selfs were included in the model as it was difficult to do when they were not separated (see differences in non-additive variance estimates in Table 2). However, it should be stated that the reliable quantification of dominance variance or inbreeding depression effect requires a large number of markers (Andrew et al. 2005). In the FS model, a selfing effect, while small, was separated from the estimated additive genetic variance and transferred to the non-additive component, resulting in the production of a heritability estimate comparable to the scenario without selfing (Table 2). All marker-based methods produced similar heritability estimates when selfs were excluded (0.06–0.07) or included (0.06–0.09), and these values are lower than the 0.10 produced by the FS pedigree model (Table 2), indicating the selfing minor role with the marker-based relationship estimation method.

Our results are similar to that reported by Blonk et al. (2010) in their commercial common sole (*Solea solea*) population. It is important to note that the accuracy of marker-based estimators is dependent on the quality (i.e., expected and observed heterozygosity, number of alleles per locus, deviation from the Hardy–Weinberg expectation, proportion of null alleles, PIC) and quantity (strict number) of the genetic markers used which are essential in unraveling

the underlying relationships needed for estimating the additive genetic variance when pedigree-free models are used (Frentiu et al. 2008). It is important to highlight the value of the marker-based method; thus, it is prudent to estimate the true relationship among the studied individuals, and this can be easily done when there is a large number (1,000 s) of markers (preferably SNPs) as they reflect the contemporary as well as the historic genealogy. A possible solution to overcome the lack of adequate markers could be the use of combination of information generated from both the marker-based relationship information and those from sibship reconstruction (Korecký et al. 2013). Moreover, marker-based relationship estimators are sensitive to populations' relatedness complexity; thus, examination of multiple estimators to select the most suitable is necessary (van de Castele et al. 2001). The advantage of the marker-based relationship is its independence from the bias associated with pedigree-based analyses as they often inflate heritability estimates due to their inability to disentangle the additive from non-additive genetic and common environmental effects, especially in long-lived forest tree populations (Sellers et al. 1998; Merilä et al. 2001; Lee et al. 2010). This was clearly demonstrated in the present study as the marker-based relationship efficiently separated the estimated additive genetic variance from its confounders (note the reduction of additive genetic variance and increase in residual variance in the marker-based models (Table 1)). This is clearly shown from the log-likelihood values which were progressively reduced from HS to FS to the marker-based models, indicating better model fit (Table 2). This trend was also mirrored but in the opposite direction where the estimated accuracy of the predicted breeding values progressively increased from HS to FS to the marker-based models (Table 2). It is important to state that when the lack of relatedness is prevalent in a population, these marker-based methods will produce biased heritability estimates due to their inability to separate genetic and environment effects (Ritland 1996; Shikano 2008).

The observed significant correlations between individuals' height breeding values across all models (Table 3) demonstrate the utility of the pedigree-free models in selection

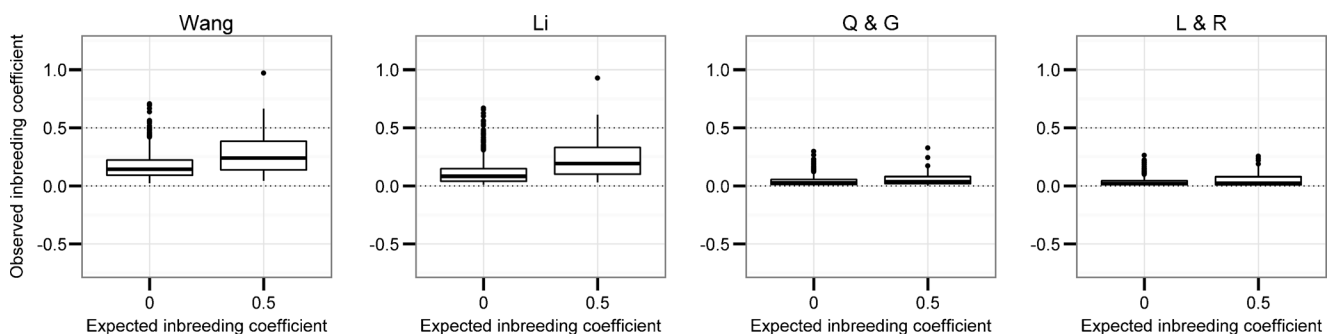


Fig. 2 Box plots portraying the distribution of marker-based inbreeding coefficient estimates within each class of expected inbreeding after nearPD function application. L & R Lynch and Ritland (1999), Q & G Queller and Goodnight (1989), Wang (2002), Li et al. (1993)

Table 2 Comparison of estimates of variance components (additive, dominance, plot, and residual), heritability ($\hat{h}^2$), log-likelihood (logL), and the estimated accuracy of the predicted breeding values (r) among western larch HS, pedigree-based FS, and four molecular genetic marker-

based pair-wise relationship estimation methods estimated using complete relationship matrix: selfing excluded and selfing included (dominance variance estimated using the fraternity matrix was possible for L & R and W; approximate standard errors are in parentheses)

Model	Additive	Dominance	Plot	Residual	Phenotypic	$\hat{h}^2$	logL	r
Selfing excluded								
HS	135.8 (64.0)	NA	23.4 (6.9)	70.4 (48.6)	229.6 (17.9)	0.59 (0.241)	−4,072.3	0.39
FS	24.0 (10.4)	0.2e−4 (0.9e−6)	43.3 (10.4)	154.1 (8.6)	221.4 (12.7)	0.11 (0.046)	−4,065.7	0.57
FS ^a	24.0 (10.4)	NA	43.3 (10.4)	154.1 (8.6)	221.4 (12.7)	0.11 (0.046)	−4,065.7	0.57
L & R	12.9 (4.2)	0.1e−4 (0.6e−6)	41.3 (9.7)	162.2 (6.6)	216.4 (11.8)	0.06 (0.019)	−4,061.6	0.85
L & R ^a	12.9 (4.2)	NA	41.3 (9.6)	162.2 (6.6)	216.4 (11.8)	0.06 (0.019)	−4,061.6	0.85
Q & G	15.9 (5.6)	NA	39.8 (9.4)	162.8 (6.6)	218.5 (11.9)	0.08 (0.025)	−4,062.4	0.88
W	13.2 (4.4)	0.4e−4 (0.2e−5)	40.1 (9.4)	162.8 (6.7)	216.1 (11.6)	0.07 (0.020)	−4,065.0	0.86
W ^a	13.2 (4.4)	NA	40.1 (9.4)	162.8 (6.7)	216.1 (11.6)	0.07 (0.020)	−4,065.0	0.86
L	14.1 (4.9)	NA	39.4 (9.3)	163.4 (6.7)	216.9 (11.7)	0.07 (0.022)	−4,064.3	0.88
Selfing included								
HS	133.8 (62.9)	NA	22.3 (6.6)	83.8 (47.7)	239.9 (17.7)	0.56 (0.228)	−4,457.4	0.44
FS	23.6 (11.4)	45.8 (14.7)	40.8 (10.0)	119.1 (14.8)	229.3 (12.6)	0.10 (0.048)	−4,445.6	0.56
FS ^a	31.1 (12.5)	NA	41.6 (10.1)	161.3 (9.5)	233.9 (13.0)	0.13 (0.051)	−4,448.6	0.62
L & R	14.8 (4.7)	4.2 (3.4)	38.4 (9.1)	169.1 (7.0)	226.6 (11.6)	0.07 (0.020)	−4,441.3	0.74
L & R ^a	15.1 (4.6)	NA	38.6 (9.1)	172.5 (6.8)	226.2 (11.6)	0.07 (0.019)	−4,442.2	0.88
Q & G	19.8 (6.4)	NA	35.7 (8.6)	122.9 (6.7)	228.5 (11.9)	0.09 (0.026)	−4,442.0	0.90
W	14.2 (4.7)	0.8 (2.3)	36.7 (8.8)	172.5 (6.9)	224.2 (11.3)	0.06 (0.020)	−4,443.1	0.77
W ^a	14.4 (4.6)	NA	36.6 (8.7)	173.1 (6.8)	224.1 (11.3)	0.06 (0.019)	−4,443.2	0.89
L	16.4 (5.3)	NA	35.1 (8.4)	173.4 (6.8)	224.9 (11.4)	0.07 (0.023)	−4,442.1	0.90

HS half-sib, FS full-sib, L & R Lynch and Ritland (1999), Q & G Queller and Goodnight (1989), W Wang (2002), L Li et al. (1993), NA not applicable

^a Models without dominance matrix

programs. The main reason for the observed differences in the correlation estimates between the marker-based methods and the FS model and those among the marker-based methods can be simply explained by what each method presents. Marker-based methods account for the Mendelian sampling term as opposed to the pedigree-based methods which represent the expected estimates in the average numerator relationship matrix (Visscher et al. 2006). Thus, the use of a pedigree-free model allows a better assessment of the within-family variance and thus maximizes the selection efficiency as both among- and with-family variations are considered during selection. Additionally, the method increases the effectiveness of the genetic resource management through better avoidance of inbreeding caused by both the buildup of co-ancestry and the loss of genetic diversity. The precision of breeding value estimates is affected not only by the number and informativeness of available markers but also by the reflection of which quantitative trait locus (QTL) alleles were transmitted from parents to offspring even when QTLs are not involved in the marker array (Zapata-Valenzuela et al. 2013). Such improvement can be achieved by construction of trait-specific relationship matrices considering the importance of

each marker to explain variance in trait under study (Zhang et al. 2011).

Finally, we would like to highlight the usefulness of the nearPD function as it paved the road to marker-based method utilization in genetic parameter estimation and selection programs. As it was mentioned above, the marker-based methods often produce a non-positive definite matrix that

Table 3 Pearson's product moment correlations of height breeding values produced from genetic analyses based on pedigree-based HS and FS and four marker-based pair-wise relationship estimation methods: selfing excluded (upper triangular) and selfing included (lower triangular)

Model	HS	FS	L & R	Q & G	W	L
HS	—	0.63	0.47	0.48	0.50	0.49
FS	0.67	—	0.60	0.60	0.63	0.63
L & R	0.49	0.64	—	0.97	0.95	0.96
Q & G	0.51	0.64	0.97	—	0.97	0.98
W	0.51	0.64	0.95	0.97	—	0.98
L	0.51	0.65	0.96	0.98	0.99	—

HS half-sib, FS full-sib, L & R Lynch and Ritland (1999), Q & G Queller and Goodnight (1989), W Wang (2002), L Li et al. (1993)

could not be utilized in linear mixed models, which the nearPD function has overcome. Moreover, the relaxing of diagonal element fixation (the difference between correlation and covariance matrices) allows inference about individuals' inbreeding levels, a situation that cannot be achieved with original relationship matrices.

Acknowledgments Financial support from the Natural Sciences and Engineering Research Council of Canada—Discovery and IRC Grants and the Johnson's Family Forest Biotechnology Endowment to YAE and funds from the National Agency for Agricultural Research (NAZV, research grant no. QJ1320013) and the Ministry of Education, Youth and Sports (KONTAKT II, research grant no. LH13021) to M. Lstibůrek are highly appreciated. The funding agencies had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Conflict of interest The authors declare that they have no conflict of interest.

Ethical standards The experiments presented in this manuscript comply with the current law of the country in which they were performed.

Data Archiving Statement Data used in this manuscript will be made publicly available through DRYAD.

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